Session 2 - Synthesis & Sampling

1. Drift

Drift is the newest instrument in Ableton, and is perfectly designed for creating classic analog sounds. As the name implies, Drift can model the inconsistent, ever-changing sound of a subtractive analog synthesizer. It is also one of the most quickly usable synths right out of the box. Before we get into that, let's take a quick look at how subtractive synthesis works.

i. Classic Sounds with Subtractive Synthesis

If you've ever used a synthesizer before, you've undoubtedly worked with **subtractive synthesis**. This type of synthesis works by creating a tone generated by an oscillator, then sculpting away at the amplitude and frequency of that tone with filters. You can imagine the process similar to a sculptor who begins with a block of marble (the tone generated by the oscillator), and chisels away at the block to get the shape they want (using the filters).

To keep our sound dynamic and interesting over time, we can use a **modulator** to change parameters over time automatically. This is most often done with an **LFO** (Low Frequency Oscillator). An LFO is just an oscillator — a looping wave of any shape — that, instead of generating a tone, is used to modulate any parameter you assign to it based on its shape.

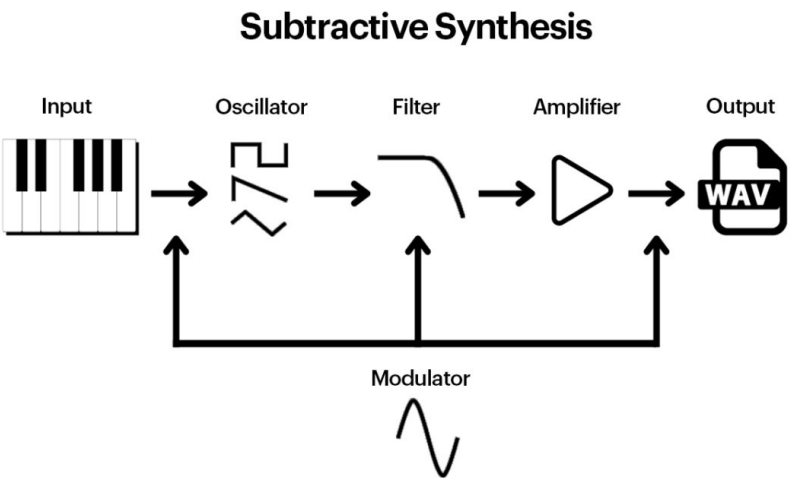

fig. 1 How sound is sculpted away with subtractive synthesis

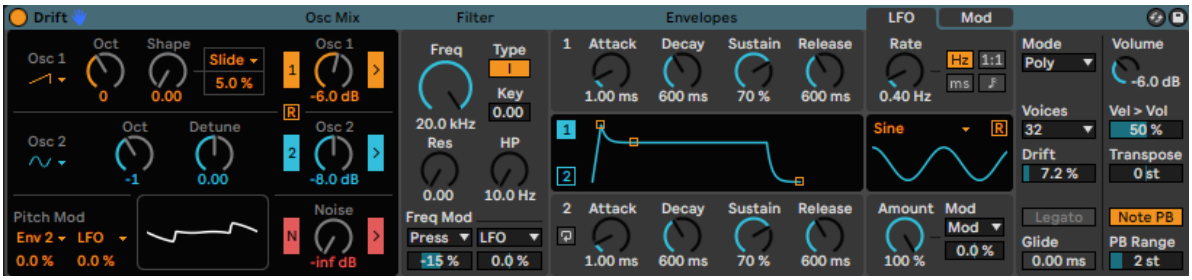

fig. 2 Drift's interface

The interface of Drift is logically laid out from left to right the same way the signal flow works in subtractive synthesis. From the left we have the "Osc Mix" with our oscillators, the filter and envelopes to sculpt the frequency and amplitude of our oscillators respectively, LFO and modulation, and finally global controls.

Drift is great for:

- Analog synth sounds
- Classic subtractive sounds
- Lo-fi & retro sounds
- Layering to create lush bass, pads, and leads

ii. Boards of Canada Lead with Drift

To demonstrate Drift's best features, let's make a quick lead that could have come from a vintage 80s synth like the Juno-60 or Jupiter-8.

To start, set both Osc 1 and 2 to saw waves in the **Osc Mix** section, and bring up the octave ("Oct") knob for Osc 2 to 0 so both oscillators are playing the same waveform at the same octave. To start introducing some warble-y sounds into the mix, set **Detune** for Osc 2 to 0.06 to detune our two waveforms very gently, similar to how an analog synth's oscillators might naturally become slightly out of tune.

Now in the **Pitch Mod** section of Osc Mix, change "LFO" to 1.5% to have our LFO slightly modify the pitch of our oscillators. In the **LFO** tab, change "Rate" to around 2.50 Hz to modulate the pitch faster.

Now in the global controls, set **Drift** to around 50%. This parameter introduces subtle changes to multiple parameters each time you press a key, so that Drift is constantly changing and evolving as you play it.

Finally, increase the "Release" to around 2.5s in the **Envelopes** tab to extend how long it plays sound after we let go of the key and thus get more of that warble-y detuned sound.

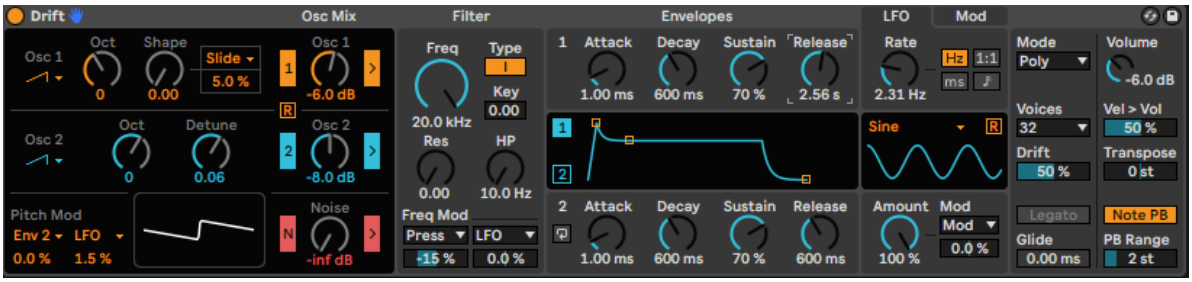

fig. 3 The finished Boards of Canada lead

From here you can increase or decrease Detune, LFO Pitch Mod, and Drift parameters to give more or less of a retro sound to your lead. Try changing Osc 1 to a pulse wave (the second option from the bottom) to get more of a late-80s FM synth sound. Reduce decay, sustain, and release values to get more of a plucky lead. Try setting **"Mode"** in global controls from Poly to Unison and increase the Unison value to around 50% to get a water-y chorused sound.

To **save your sound as a preset**, click the floppy disk icon on the very top right of your instrument. This will create a new .adv file in the User Library in the File Browser that you can rename to whatever you like. To later load this preset, just navigate to the instrument (in this case, Drift) in the User Library and double click the preset. To save a preset downloaded elsewhere, such as the ones in the course drive, navigate to Ableton\User Library\Presets\Instruments and save the preset in the instrument folder.

2. Operator

Operator is one of the most flexible and powerful instruments in Ableton, but it can often be intimidating on first glance. What makes Operator so powerful? Primarily its multiple forms of sound synthesis.

Operator is great for:

- Bass (sub, reese, tech, garage, etc.)
- Metallic, bell-like pads and leads
- Quick simple subtractive synths
- SFX, noise, risers
- Digital off-kilter timbres

i. Clicky Kicks with Operator

Operator is excellent for quickly designing percussion. Simply load up a new Operator, and set “Initial” in the Envelope section of Oscillator 1 to 0 dB. Increase “Attack” to around 20 ms, Decay to around 180 ms, Sustain to -inf dB, and Release to 50ms.

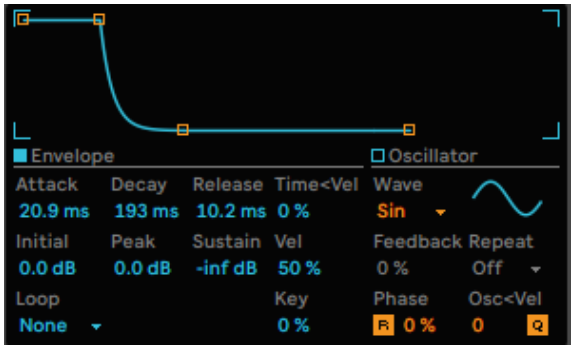

fig. 4 Oscillator 1 Envelope

Now turn on the **Pitch Envelope** by clicking the toggle and increase the “Pitch Env” knob to 100% so it is at full effect. This pitch envelope will add a quick increase or decrease in pitch to our sound depending on how we set it, and is great for getting punchy or clicky transients. Finally, shape the pitch envelope by increasing “Peak” all the way to +48 st and Decay down to around 50 ms. Now just play a note around C1 to D2 and you have a kick that will cut through any mix. Next week we’ll explore some effects devices that can be used to greatly increase the fatness and punchiness of this kick.

ii. Mathematical Chaos with FM Synthesis

Operator is great for quickly getting classic synth and percussion sounds — but what if we want to explore more interesting sonic territory? Enter: FM Synthesis.

FM (Frequency Modulation) Synthesis works by having one oscillator, the **modulator wave** control the frequency of another oscillator, the **carrier wave**. This works similar to the modulator we saw in subtractive synthesis, but at a much higher rate.

This interaction produces very complex waveforms that can often sound atonal if we don’t use whole number (integer) ratios between the two oscillators frequencies. For example, if our modulator frequency is 800Hz and our carrier wave is 400Hz, we have a ratio of 2:1. This will create a complementary **overtone**, or frequency that sits above our primary tone at 400Hz.

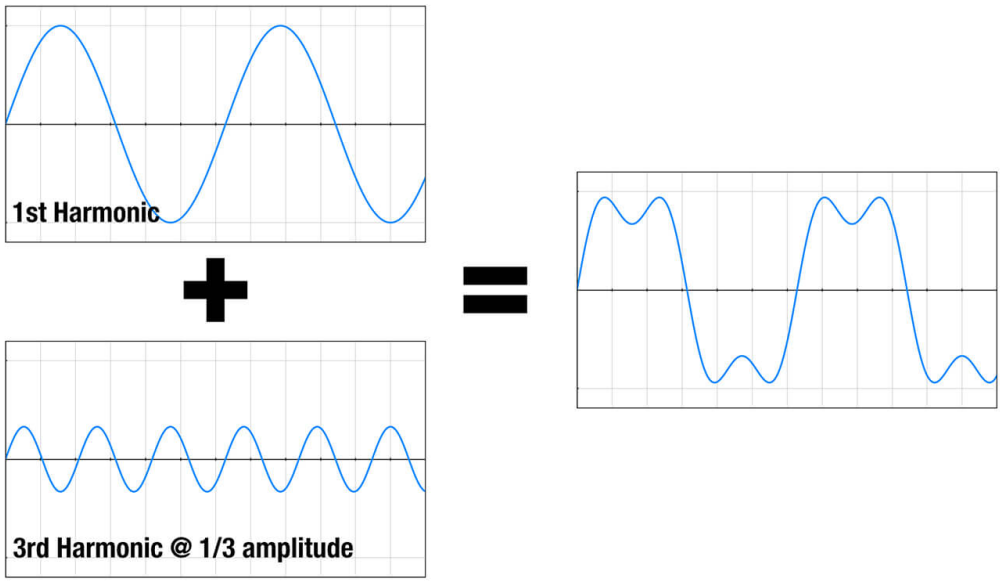

fig. 5 How FM synthesis works. The top wave (”1st Harmonic”) is the carrier wave, and the bottom (”3rd Harmonic”) is the modulator wave. Added together, we get the more complex waveform on the right

Moving out of these integer ratios, such as a ratio of 2:1.25 introduced grittier more atonal textures. While these can quickly get out of control, we can harness these uneven overtones to create all kinds of interesting textures and timbres. Because of this, Operator’s FM capabilities make it a powerful tool for creating aggressive basses, punchy kicks, and a whole array of strange sounds. To demonstrate this, let’s start with a plucky FM bass with Operator.

Plucky FM Bass with Operator

When you first load Operator, you’ll only have a single oscillator audible, oscillator A. This is because it’s the only oscillator with its “Level” parameter audible.

However, Operator also loads up already in FM mode, as we can see from the vertical diagram of colored boxes in the bottom left of the global controls. Changing this can change Operator into a number of different synthesis modes.

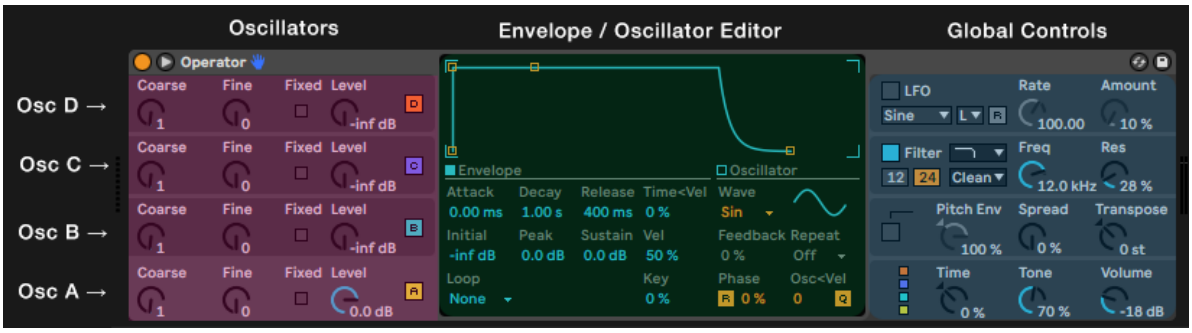

fig. 6 Operator's main control panels

To get FM synthesis going, we need to bring the Level control of oscillator B up. Bring it up to -10dB. If you play some notes, you'll already hear a nice plucky bass. With Oscillator B highlighted, bring down the sustain and increase decay of its envelope in the envelope editor to give it a quieter pluck.

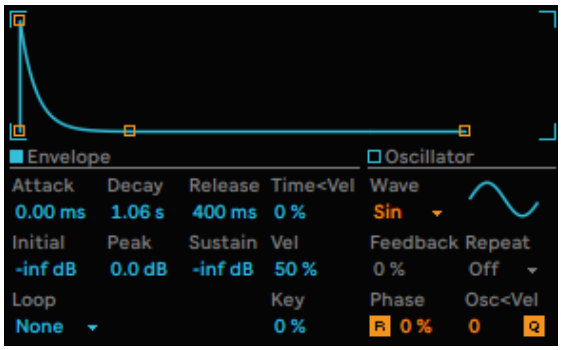

fig. 7 The envelope for Oscillator B

Detune Oscillator B by an octave by setting its Coarse to 0.5. Now to get some more power in our bass, we need to use a more aggressive waveform than just a sine wave. With Oscillator B selected, change Wave under the Oscillator tab of the Envelope/Oscillator Editor from **sine** to **square 4** — a square wave with 3 harmonics.

Highlight Oscillator A and play around with the decay and sustain to dial in how plucky you want your bass to be.

Finally, go to the global controls in the bottom right, and set "Voices" to 1 to turn our bass from polyphonic (multiple notes can be heard at once) to monophonic (only one note can be heard at once), and turn the "Volume" up until comfortable. This prevents overlapping notes from clashing and creating mud.

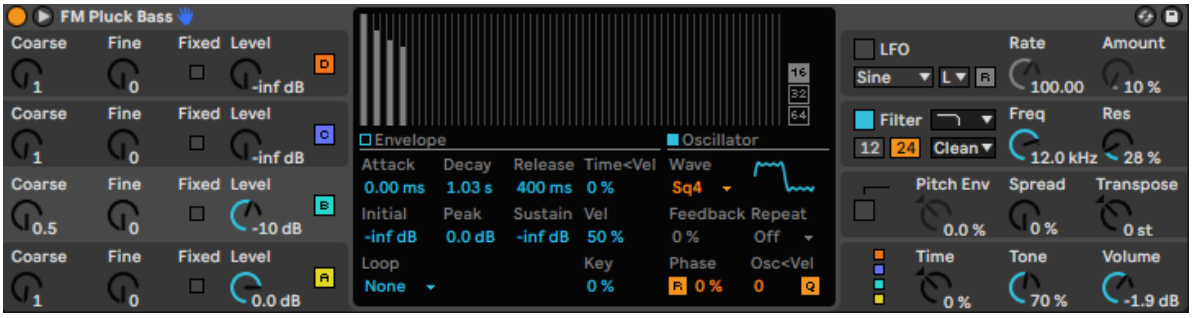

fig. 8 The finished FM pluck bass

This is just the beginning. To get more interesting, aggressive, or strange sounds, try playing around with **"Fine"** knobs on each oscillator to detune them. See what happens if you introduce Oscillators C and/or D by turning up their Level knobs. In the Oscillator waveform, try clicking and dragging in new harmonics with the **pencil tool** to introduce enharmonic frequencies into the sound. With Operator you can make any sound as musical or non-musical as you like.

3. Wavetable

Wavetable, as its name implies, uses **wavetable synthesis**. This type of synthesis works by storing multiple waveforms (like the sine or saw waves we used in Operator) within a single **table**, and allowing you to morph and scan between these stored waveforms over time. This can produce very dynamic and organic sounds with an endless possibility of exciting timbres, as we can use any waveforms we want within our table. This form of synthesis is also found in many popular VST synths such as Serum, Vital, and Pigments.

Wavetable is great for:

- Pads
- Ambient SFX
- Leads and basses that morph over time

Evolving Pad with Wavetable

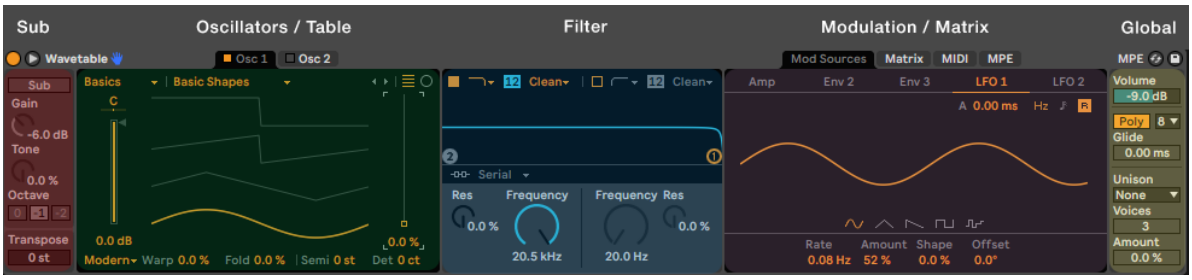

fig. 9 The five sections of wavetable

Wavetable can be broken into 5 sections. From left to right these are:

1. **Sub** - Controls for a sub oscillator
2. **Oscillators / Table** - Control of the table for each of the two oscillators
3. **Filter** - A classic circuit for the oscillators
4. **Modulation / Matrix** - Amplitude control, modulation matrix, MIDI mapping, and MPE control
5. **Global** - Controls for global volume, play style, and unison

When you load Wavetable, it loads up with the “Basic Shapes” table selected, which contains a sine, triangle, saw, and square wave. To scan through these, we can use the wavetable position slider along the right side of the table. Leave it around 35%. This is great for getting a simple subtractive sound, but to unlock the true potential of Wavetable we need to go into the **Matrix**.

The Matrix contains a grid of target parameters on the left, and modulation sources on the top right. By clicking and dragging in one of these intersecting squares, we can have the corresponding modulation source on the top modulate the parameter on the left. Let’s try this by having LFO 1 modulate Osc 1 Pos by 50. Play a note, and you’ll see the wavetable position changing.

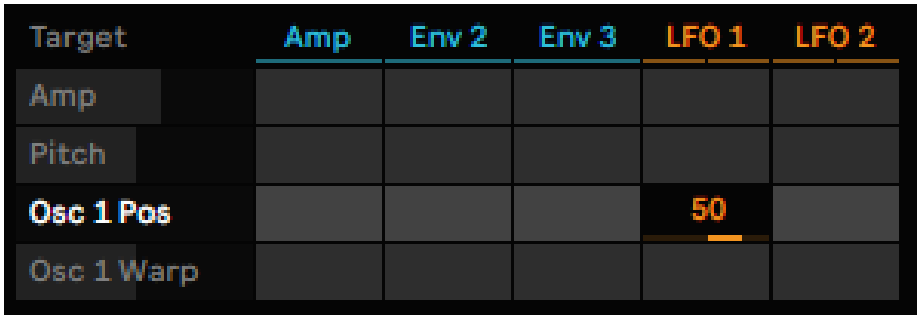

fig. 10 LFO 1 is effecting the table position of oscillator 1 by 50%

Click on LFO 1 to open its controls. Set the “Rate” to something slower, like 0.08 Hz, and bring the “Amount” down to around 60%. We can now hear the wavetable smoothly and slowly being scanned through, producing a dynamic sound as we hold a note or chord.

The fastest way to get something beautiful out of Wavetable, and my personal favorite feature, is Wavetable’s **Unison** modes in the Global controls. Want an Arca-style chorus-y pad? Simply load up a put Unison on “Shimmer” with “Amount” set to 75-80%. What about something more aggressive and buzzing like an EDM pad? use “Noise” mode with “Amount” at 30-50%. For our purposes, let’s use “Shimmer” with 3 voices at 45%.

fig. 11 Unison settings for an evolving pad

Next, let’s move over to the filter section and bring our high cut filter down to around 9 Hz with a resonance of 35% to smooth out our sound. Click on “Amp” under Mod Sources and change Attack to 1.00s and Sustain to 0.0 dB to get a nice soft pad.

To get a fuller, deeper sound, turn on Oscillator 2 by clicking the small gray box next to Osc 2 above the table. Set the wave to “Sub 1” with the dropdown, and set “Semi” to -12 st to detune oscillator 2 by an octave. Lower the volume of oscillator 2 to -6 dB using the slider on the left of the table, and move its wave position to 30%.

To get some modulation on our second oscillator, open the Matrix tab, then simply click the wave position for oscillator 2, and you’ll see “Osc 2 Pos” appear as the last target in the matrix. Set LFO 2 to control it by 35%, then go in to LFO 2 and set the Rate to 0.12 Hz. Now simply dial in the wave position, filter, and modulation amounts to taste.

This type of pad is best played with sparse extended chords (e.g: root-5th-7th [such as C2 G2 D3], root-7th-9th [such as D2, C3, E3], etc.)

Endless Timbres with Custom Wavetables

While Wavetable offers a good diversity of waves built-in, the real potential of Wavetable is unlocked when we utilize the fact that Wavetable will let us use any sound as a wavetable.

To do so, simply drag and drop any sound over the table and Wavetable will automatically load it in. The possibilities here are endless.

Some ideas to try:

- Use a highly textured sound, such as glass being crunched, metal being scraped, leather/latex/plastic being stretched, etc. and detune it all the way by setting semi to -24 st. Now play a low note (if using your computer keyboard to play, press Z and X to shift octaves) for a highly textured aggressive bass.
- Resample a synth by recording a single note from the original synth and using it as a wavetable. Try tweaking parameters on the original synth and recording it again, then using this as the wavetable for Osc 2 and adjust the gain accordingly. This lets you effectively use Wavetable as a complex sampler.
- Use the **Warp** and **Fold** parameters at the bottom of the wavetable viewer to stretch and warp your custom wavetables. Warp is particularly effective at pulling out textures from sounds, while fold can produce noisy FM-like sounds at higher values.

4. Generative Composition with Follow Actions

Let's take a break from exploring new instruments to explore one of the most underappreciated features of Ableton and the coolest part of using session view: **Follow Actions.**

Follow actions change the way that Ableton switches between clips when triggered. By default when we switch between clips, Ableton will wait until the next bar (or whatever setting you set in the Quantization Menu in the toolbar), then switch to the next selected clip.

With follow actions, we can tell Ableton to behave very differently when we trigger a clip. To start, activate the Follow toggle in the toolbar.

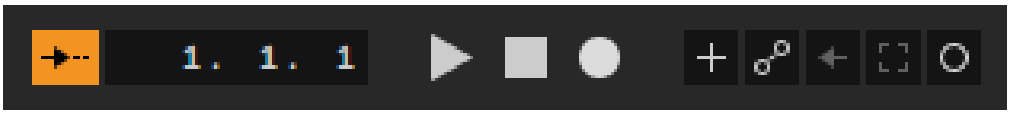

fig. 12 The follow button is on the far left

Next we need to have some clips to play around with. Let's use MIDI. Create a new MIDI track with an instrument of your choice in it, and create two new clips. In one, create a 1 bar long note at B2 (hint: a quick way to do this is to double click to add a note, then click the legato button to have that note fill out the entire 1 bar long clip). Do the same thing on the same MIDI track with C#3 and D3.

fig. 13 A MIDI track with three clips added

Notice how the play buttons for these clips are striped — this indicates that follow actions are on.

Click on B open the MIDI Note Editor, then highlight all three clips by clicking on one, then hold shift and press ↓. Now go to the Follow Actions tab in the clip menu and enable Follow Action. Below this button, click the dropdown menu “Next” and change it to **“Other”**. Activate any of our three clips and notice how the clips are randomly being switched between on loop.

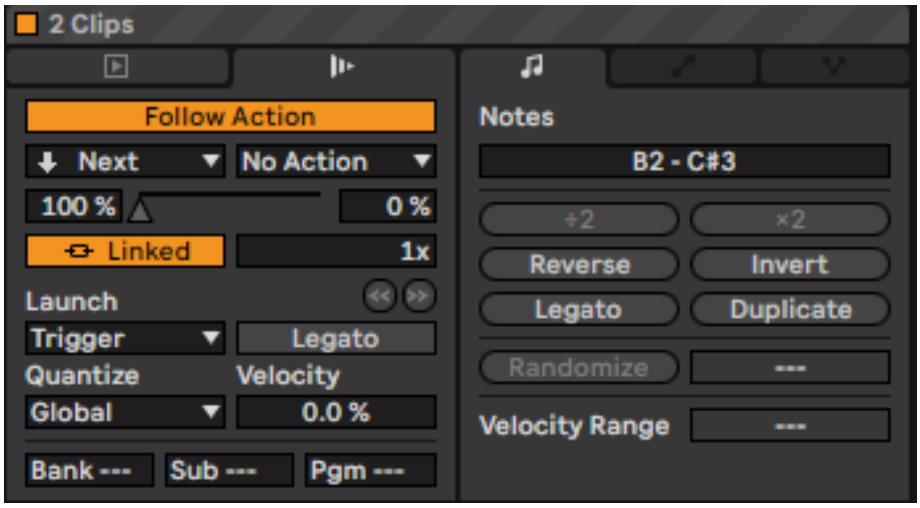

fig. 14 Follow Actions enabled

If we can do this with just three notes, let's see what we can do when we fill out the notes of an entire scale. We already have the first three notes of B Minor (B, C#, D), so let's add the remaining 4 clips we need to get the full scale: E3, F#3, G3, and A3). To keep the follow action settings we just added to these new notes, you can simply copy one of the notes we already have by clicking and dragging it down to the next empty clip slot in our MIDI track while holding CTRL. Now just change the note and name of this new clip.

Now as we play a clip, we begin generating a melody using any notes in B minor. However, these notes are currently all one bar long.

To get more variety in our note lengths, highlight all clips again and in the follow actions tab, untoggle the **"Linked"** button. Linked mode means that each new clip is only triggered at the end of the previous clip. The toggle now becomes **"Unlinked"** and a note length dialog will appear next to it. This now allows us to switch to a new clip before the end of the previous, and the note length dialog lets us choose how often we want these notes to change. Let's try setting it to 8th notes (0.0.2).

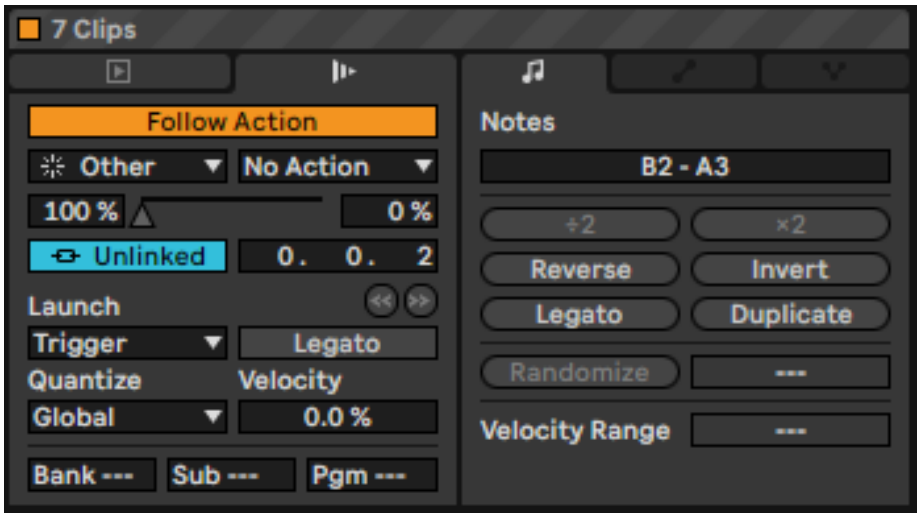

fig. 15 Unlinked 8ths notes

From here, you can play around with the attack and release settings on your instrument to taste, and experiment with different unlinked note lengths, as well as the global BPM.

Once you've dialed in the sound that you like, you can save the composition by recording the generated MIDI into another MIDI track. Create a new MIDI track, then select the generative MIDI track we've been working with under "MIDI From". Arm the new MIDI track, then just hit the record button in a blank MIDI clip and it will begin capturing the generated MIDI.

fig. 16 Recording from the first MIDI track into the second

From here you can get more randomized variation by using the MIDI Note Editor to reverse, invert, shorten/lengthen, and duplicate sections of notes. Want a generative chord progression? Instead of individual notes, place clips of each chord you want to use instead. You can also record to audio, then use some of the warp mode techniques we learned in session 1 to create entirely new sounds without having to play a single note ourselves!

5. Sampler

Granular Synthesis with Sampler

Granular synthesis is, simply put, my favorite sound in electronic music. It is both instantly recognizable and infinitely interesting, limited only by the content you put into it. It is the signature sound of much contemporary electronic music, and has been the backbone of multiple experimental genres, from early-2000s glitch to contemporary leftfield electronic music.

Granular synthesis is actually a special kind of sampling technique that is functionally similar to wavetable synthesis. While a normal sampler, such as Ableton's Sampler device, treats your sample as a single piece, a granular synthesis breaks your sample up into tons of extremely tiny pieces called **grains**.

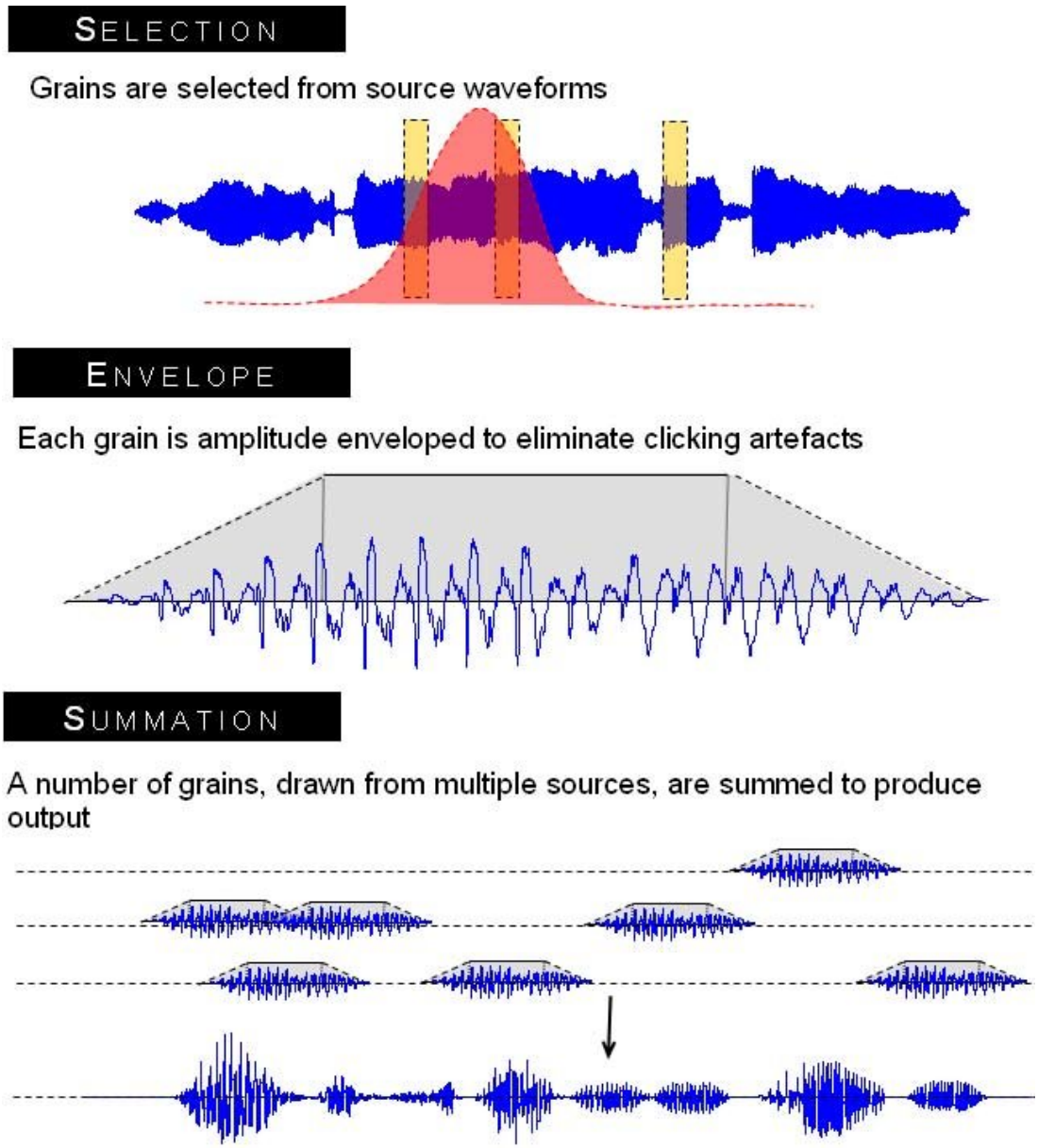

fig. 17 The process of sampling within granular synthesis

A granular synthesizer's sound is shaped primarily by controlling how big these grains are (**Grain Size**), where in the sample it is playing these grains from (**File Position**), and how much it'll jump around to neighboring grains as it plays (**Spray**). With these parameters we can get anything from oblique choppy chaos to beautiful lush pads and everything in between.

While there are many dedicated granular synthesizers out there (I highly recommend the free Max for Live device, Granulator II, in particular), we can use Sampler's high degree of control to effectively turn it into a powerful granular synthesizer.

To start, we need to open up a Sampler and load in a sample. Any sample will do, but think about your end goal when picking your samples. Want a smooth tonal pad? Pick a longer sample with a strong fundamental frequency. Want something strange and chaotic? Use an atonal sample with a lot of different transients to jump around to. In this case, we'll use a sample of some wind chimes as an example.

fig. 18 Sampler with wind chimes loaded as a sample

Press a key and you'll see the playhead start playing your sample from the Sample Start (gray flag on the far left of the waveform) until the Sample End (gray flag on the right). To get Sampler to loop as we hold down a key, toggle the button under "Sustain Mode" underneath the waveform with two forward arrows. **Loop brackets** have been added as indicated by the blue highlight in the waveform.

To start getting a granular sound, click and drag the "Loop End" parameter from the menu or waveform to the left until it nearly meets the "Loop Start". Click and drag in the waveform to zoom in to our loop brackets, and bring them close enough together that a new pitch begins to form. Click and drag the top bar of the loop bracket to move your loop to a louder section of the sample. Bring the Sample Start near this new position so it will play near the loop every time you play a new note.

fig. 19 A tiny section of the sample selected with the loop parameters

To reduce clicks, drag up the “Crossfade” parameter so the playhead fades the volume in and out each time it loops.

With a small enough grain size, we can effectively get a distinct pitch to our sample that gets higher with a smaller grain size. To get perfect tuning, go into Audio Effects in the File Browser, then drag and drop a **Tuner** device on the MIDI track with your Sampler. This device will show you what pitch Sampler is producing, allowing you to dial in the “Loop End” parameter to a perfect pitch of your choosing.

Try clicking “Loop End” to highlight it, then use the up and down arrow keys on your keyboard while playing a note to dial in an exact pitch that at least shows up green on Tuner. Now set the “Root” parameter in Sampler to this pitch (in my case, A3). This will now properly pitch your Sampler so that it corresponds to the keys on a keyboard. As long as we keep this loop length the same, it will always be in pitch as we move around our sample.

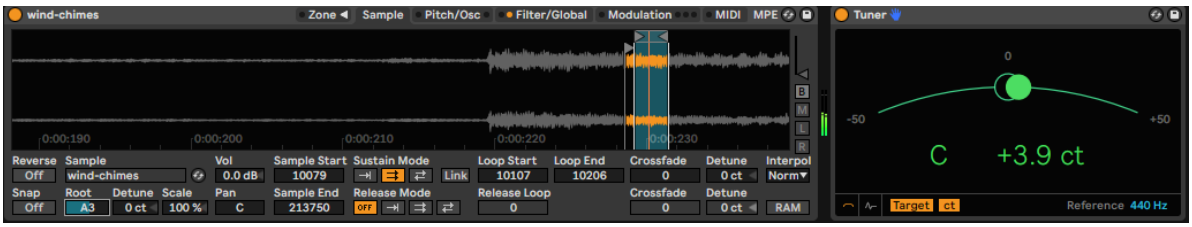

fig. 20 Properly pitching our sampler by setting Root to A3 using Tuner

This is great to start with, but to get more dynamics, we need to modulate where these grains are playing in our sample. First, turn on “Release Mode” by clicking the second toggle in the Menu (→|). This lets us modify our loop placement. Next, go to the

“Modulation” tab in the top of Sampler. Here we have an array of LFOs that we can map to any parameter we want. Activate LFO 2 by clicking its corresponding toggle, then find the targets at the bottom labeled “A” and “B”. Select “Loop Start” in the dropdown menu next to “A”. Tell Sampler we want it to fully effect the parameter by setting the number box next to it to 100.00. Now set the LFO shape to a **Sample & Hold** function (the last option under “Type”) to allow the parameter to randomly change and hold its position around our Sample. Finally, set the frequency to taste — in our case: 3.36 Hz. Now play a note, and notice the playhead jumping around in the waveform. Try playing with different waveforms to have the loop instead scan smoothly through the sample for a nice pad sound, or a high frequency sample & hold for something more chaotic.

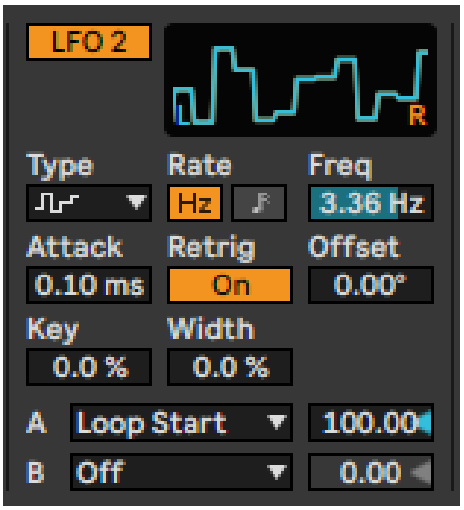

fig. 21 LFO 2 modulating Loop Start

What makes Sampler so powerful over many dedicated granular synthesizers is its huge number of envelopes, filters, and modulation sources. We can add some **FM** to our sound by going to the Pitch/Osc tab, turning on the top envelope by clicking the “Osc” toggle, then bringing up the “Volume” control on the right. Try shaping the envelope by bringing “Sustain” up to 0.0 dB and “Release” to 1.00 ms, then play around with the Volume control.

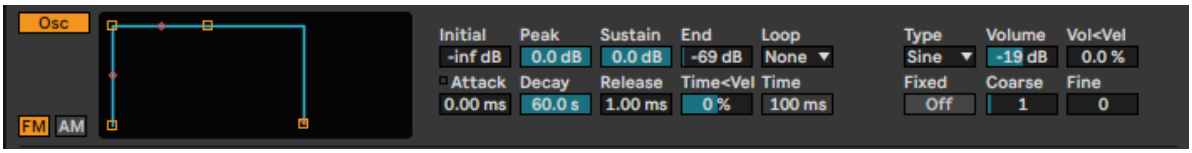

fig. 22 FM added to the granular synthesizer with the Osc Envelope

We can use the **Pitch Envelope** below the Osc to add punchiness to our sound by turning on the “P. Env” toggle, and setting “Amount” to a high value (usually +20 - +48 st works well). Play around with the “Decay” parameter to increase or decrease how long the envelope takes to get back to its original pitch. Wildly changing values in “Transp” can lead to interesting sounds as well.

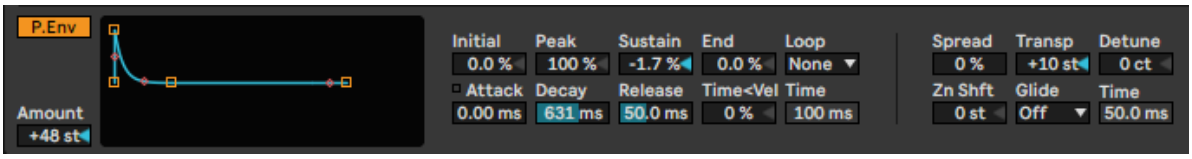

fig. 23 Adding punch and pitch modulation with the Pitch Envelope